Lecture 4: Linear Algebra Foundations

Numerical Computing

Francisco Richter Mendoza
Università della Svizzera Italiana

September 10, 2025

Overview

Linear algebra provides the mathematical foundation for a vast array of numerical methods. This lecture develops the theoretical framework necessary for understanding numerical linear algebra, focusing on the mathematical structures and properties that determine the behavior of computational algorithms. We explore vector and matrix norms, analyze special classes of matrices, and introduce the singular value decomposition as a unifying tool for understanding matrix properties.

The transition from finite-dimensional linear algebra to numerical computation introduces new considerations. While theoretical linear algebra often assumes exact arithmetic, numerical methods must account for roundoff errors, finite precision, and the computational complexity of operations. This perspective leads us to emphasize concepts such as condition numbers, matrix norms, and stability analysis.

Our approach emphasizes the mathematical theory underlying numerical linear algebra rather than computational procedures. We develop tools for analyzing the sensitivity of linear algebra problems to perturbations, understanding the geometric properties of matrices, and characterizing the fundamental limitations of numerical computation. These theoretical insights provide the foundation for the direct and iterative methods we will study in subsequent lectures.

---

1 Vector and Matrix Norms

1.1 Vector Norms

Definition. Vector Norm

A function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ is a **vector norm** if it satisfies:

1. **Positivity:** $\|x\| \geq 0$ for all $x \in \mathbb{R}^n$, with $\|x\| = 0$ if and only if $x = 0$
2. **Homogeneity:** $\|\alpha x\| = |\alpha|\|x\|$ for all $\alpha \in \mathbb{R}$, $x \in \mathbb{R}^n$
3. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in \mathbb{R}^n$

Definition. Common Vector Norms

For $x = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$:

p-norms: $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p}$ for $1 \leq p < \infty$

Specific cases:

- $\|x\|_1 = \sum_{i=1}^n |x_i|$ (Manhattan norm)
- $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$ (Euclidean norm)
- $\|x\|_\infty = \max_{1\le i\le n} |x_i|$ (Maximum norm)

Theorem 1.1 (Norm Equivalence)**.** *All norms on $\mathbb{R}^n$ are equivalent. Specifically, for any two norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ on $\mathbb{R}^n$, there exist positive constants c_1 and c_2 such that:*

$$c_1\|x\|_\alpha \le \|x\|_\beta \le c_2\|x\|_\alpha$$

for all $x \in \mathbb{R}^n$.

Important inequalities:

$$\|x\|_\infty \le \|x\|_2 \le \sqrt{n}\|x\|_\infty \tag{1}$$
$$\|x\|_2 \le \|x\|_1 \le \sqrt{n}\|x\|_2 \tag{2}$$
$$\|x\|_\infty \le \|x\|_1 \le n\|x\|_\infty \tag{3}$$

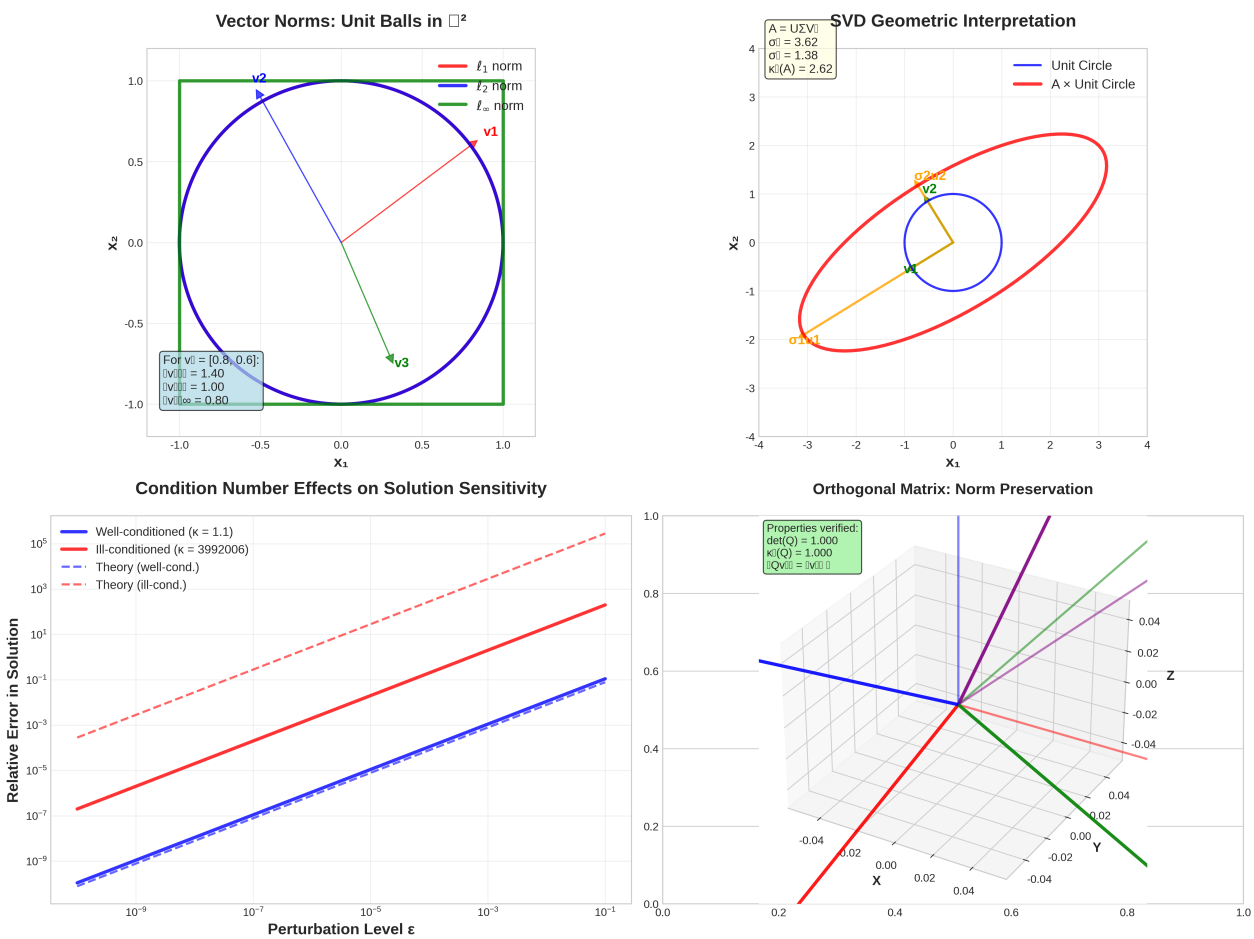

Figure 1: Linear algebra foundations visualization. Top left: Unit balls for different vector norms (ℓ_1, ℓ_2, ℓ_∞) showing their geometric properties and sample vector norm calculations. Top right: SVD decomposition components illustrating the geometric interpretation with singular vectors and values. Bottom left: Matrix condition number effects on linear system sensitivity with well-conditioned vs ill-conditioned examples. Bottom right: Orthogonal matrix properties demonstrating norm preservation and geometric structure.

1.2 Matrix Norms

Definition. Matrix Norm

A function $\|\cdot\| : \mathbb{R}^{m\times n} \to \mathbb{R}$ is a **matrix norm** if it satisfies the vector norm axioms. Additionally, for square matrices, we often require:

Submultiplicativity: $\|AB\| \leq \|A\|\|B\|$ for compatible matrices A and B.

Definition. Induced Matrix Norms

Given a vector norm $\|\cdot\|$ on $\mathbb{R}^n$, the **induced matrix norm** (or operator norm) is:

$$\|A\| = \max_{x\neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|$$

Common induced norms:

- $\|A\|_1 = \max_{1\leq j\leq n} \sum_{i=1}^{m} |a_{ij}|$ (maximum column sum)
- $\|A\|_\infty = \max_{1\leq i\leq m} \sum_{j=1}^{n} |a_{ij}|$ (maximum row sum)
- $\|A\|_2 = \sigma_1(A)$ (largest singular value)

Definition. Frobenius Norm

The **Frobenius norm** is defined as:

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} |a_{ij}|^2} = \sqrt{\mathrm{tr}(A^T A)} = \sqrt{\sum_{i=1}^{\min(m,n)} \sigma_i^2}$$

where σ_i are the singular values of A.

The Frobenius norm is not an induced norm but satisfies submultiplicativity.

Theorem 1.2 (Properties of Induced Norms)**.** *For an induced matrix norm $\|\cdot\|$:*

1. $\|I\| = 1$ *(where I is the identity matrix)*
2. $\|Ax\| \leq \|A\|\|x\|$ *for all compatible A and x*
3. $\|AB\| \leq \|A\|\|B\|$ *for compatible matrices A and B*
4. *If A is invertible, then* $\|A^{-1}\| \geq 1/\|A\|$

Example 1.1 (Norm Calculations)**.** *Consider the matrix* $A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$. *We can compute:*

```python
import numpy as np
A = np.array([[3, 1], [1, 2]])

# Different matrix norms
norm_1 = np.linalg.norm(A, ord=1)       # ||A||_1 = 4
norm_inf = np.linalg.norm(A, ord=np.inf)  # ||A||_inf = 4
norm_2 = np.linalg.norm(A, ord=2)       # ||A||_2 approx 3.73
norm_F = np.linalg.norm(A, ord='fro')   # ||A||_F approx 3.87

print(f"||A||_1 = {norm_1}")
print(f"||A||_inf = {norm_inf}")
print(f"||A||_2 = {norm_2:.2f}")
print(f"||A||_F = {norm_F:.2f}")
```

The 2-norm equals the largest singular value: $\sigma_1 \approx 3.73$.

---

2 Condition Numbers and Sensitivity Analysis

2.1 Matrix Condition Numbers

Definition. Condition Number

For an invertible matrix $A \in \mathbb{R}^{n \times n}$ and a given matrix norm $\|\cdot\|$, the **condition number** is:

$$\kappa(A) = \|A\|\|A^{-1}\|$$

For the 2-norm: $\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}$
where $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of A.

Theorem 2.1 (Properties of Condition Numbers)**.**

1. $\kappa(A) \geq 1$ *for any invertible matrix* A
2. $\kappa(I) = 1$ *(identity matrix is perfectly conditioned)*
3. $\kappa(\alpha A) = \kappa(A)$ *for any* $\alpha \neq 0$
4. $\kappa(A) = \kappa(A^{-1})$
5. *For orthogonal matrices* Q*:* $\kappa_2(Q) = 1$
6. $\kappa(AB) \leq \kappa(A)\kappa(B)$

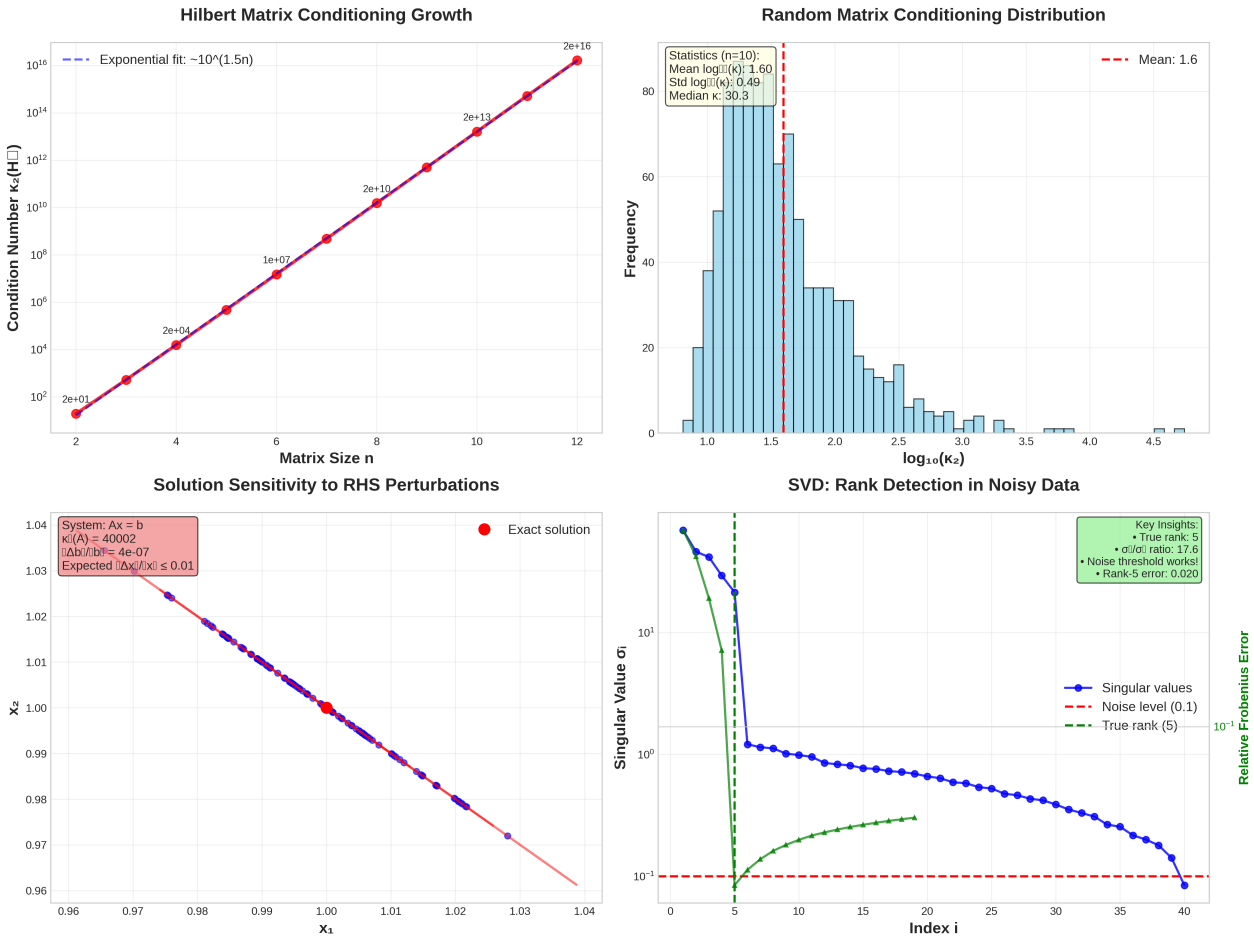

Figure 2: Matrix conditioning analysis. Top left: Hilbert matrix conditioning growth showing exponential increase with matrix size. Top right: Random matrix conditioning distribution with statistical analysis. Bottom left: Perturbation analysis visualization showing solution sensitivity to right-hand side perturbations with error ellipses. Bottom right: SVD rank detection in noisy data demonstrating singular value decay and approximation quality.

2.2 Perturbation Analysis for Linear Systems

Theorem 2.2 (Perturbation Bound for Linear Systems)**.** *Consider the linear system $Ax = b$ where A is invertible. If we perturb the right-hand side to $b + \Delta b$, the solution changes from x to $x + \Delta x$ where $A(x + \Delta x) = b + \Delta b$.*

Then:

$$\frac{\|\Delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\Delta b\|}{\|b\|}$$

This shows that the condition number bounds the relative error amplification.

Theorem 2.3 (Perturbation in the Matrix)**.** *If we perturb the matrix: $(A + \Delta A)(x + \Delta x) = b$, and $\|\Delta A\| < 1/\|A^{-1}\|$, then:*

$$\frac{\|\Delta x\|}{\|x + \Delta x\|} \leq \frac{\kappa(A)\|\Delta A\|/\|A\|}{1 - \kappa(A)\|\Delta A\|/\|A\|}$$

For small perturbations: $\frac{\|\Delta x\|}{\|x\|} \approx \kappa(A) \frac{\|\Delta A\|}{\|A\|}$

Example 2.1 (Hilbert Matrix Conditioning)**.** *The Hilbert matrix H_n with entries $h_{ij} = 1/(i + j - 1)$ is notoriously ill-conditioned:*

```python
from scipy.linalg import hilbert
import numpy as np

# Generate Hilbert matrices of different sizes
for n in range(2, 11):
    H = hilbert(n)
    kappa = np.linalg.cond(H)
    print(f"H_{n}: kappa_2 = {kappa:.2e}")

# Output shows exponential growth:
# H_2: kappa_2 = 1.93e+01
# H_3: kappa_2 = 5.24e+02
# H_4: kappa_2 = 1.55e+04
# H_5: kappa_2 = 4.77e+05
# ...
# H_10: kappa_2 = 1.60e+13
```

For $n = 10$: $\kappa_2(H_{10}) \approx 1.6 \times 10^{13}$. This means that a relative perturbation of 10^{-16} in the data could result in a relative error of 10^{-3} in the solution, making the solution meaningless in double precision arithmetic.

Intuition: Geometric interpretation of conditioning

The condition number measures how "close" a matrix is to being singular:

- **Well-conditioned** ($\kappa \approx 1$): Matrix transformation preserves relative distances
- **Ill-conditioned** ($\kappa \gg 1$): Matrix transformation severely distorts the geometry
- **Singular** ($\kappa = \infty$): Matrix collapses the space to a lower dimension

The condition number quantifies the maximum distortion factor for any vector under the linear transformation. In the context of linear systems, it measures how small changes in the input (right-hand side or matrix entries) can be amplified in the output (solution vector).

---

3 Special Classes of Matrices

3.1 Orthogonal Matrices

Definition. Orthogonal Matrix

A matrix $Q \in \mathbb{R}^{n\times n}$ is **orthogonal** if $Q^TQ = I$, equivalently $Q^T = Q^{-1}$.

For rectangular matrices $Q \in \mathbb{R}^{m\times n}$ with $m \geq n$, we say Q has **orthonormal columns** if $Q^TQ = I_n$.

Theorem 3.1 (Properties of Orthogonal Matrices)**.** *If Q is orthogonal, then:*

1. $\|Qx\|_2 = \|x\|_2$ *for all x (preserves Euclidean norm)*
2. $\det(Q) = \pm 1$
3. *All eigenvalues have absolute value 1*
4. $\kappa_2(Q) = 1$ *(perfectly conditioned)*
5. $(Qx)\cdot(Qy) = x \cdot y$ *for all x, y (preserves inner products)*

3.2 Symmetric Matrices

Definition. Symmetric Matrix

A matrix $A \in \mathbb{R}^{n \times n}$ is **symmetric** if $A = A^T$.

Theorem 3.2 (Spectral Theorem for Symmetric Matrices)**.** *Every symmetric matrix A can be diagonalized by an orthogonal matrix:*

$$A = Q\Lambda Q^T$$

where Q is orthogonal and $\Lambda = diag(\lambda_1, \lambda_2, \ldots, \lambda_n)$ with real eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$.

Moreover: $\kappa_2(A) = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$ (assuming A is invertible).

Definition. Positive Definite Matrices

A symmetric matrix A is **positive definite** if $x^T Ax > 0$ for all $x \neq 0$.

Equivalent conditions:

- All eigenvalues are positive
- All leading principal minors are positive
- Cholesky decomposition $A = LL^T$ exists with L lower triangular

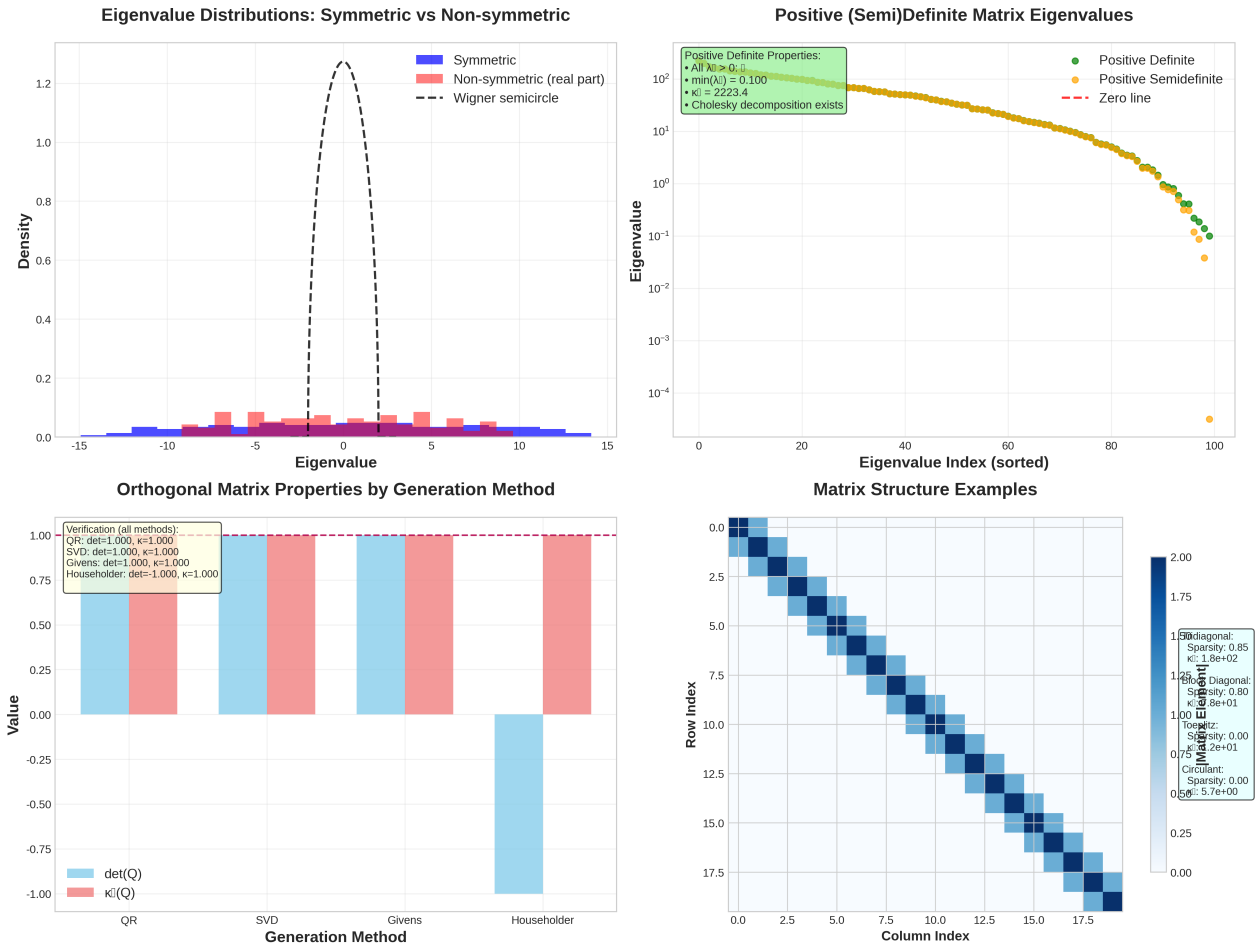

Figure 3: Special matrix classes properties. Top left: Symmetric matrix eigenvalue distributions compared to non-symmetric matrices with Wigner semicircle law approximation. Top right: Positive definite vs positive semidefinite eigenvalue comparison. Bottom left: Orthogonal matrix generation methods verification showing determinants and condition numbers. Bottom right: Matrix structure examples including tridiagonal, block diagonal, Toeplitz, and circulant matrices with sparsity analysis.

---

4 Singular Value Decomposition

4.1 SVD Theorem and Properties

Theorem 4.1 (Singular Value Decomposition)**.** *Every matrix $A \in \mathbb{R}^{m\times n}$ has a singular value decomposition:*

$$A = U\Sigma V^T$$

where:

- $U \in \mathbb{R}^{m\times m}$ *is orthogonal (left singular vectors)*
- $V \in \mathbb{R}^{n\times n}$ *is orthogonal (right singular vectors)*
- $\Sigma \in \mathbb{R}^{m\times n}$ *is diagonal with* $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min(m,n)} \geq 0$

The σ_i are called ***singular values*** *of A.*

Theorem 4.2 (SVD Properties)**.** *For $A = U\Sigma V^T$:*

1. $\|A\|_2 = \sigma_1$ *(largest singular value)*
2. $\|A\|_F = \sqrt{\sum_{i=1}^{\min(m,n)} \sigma_i^2}$
3. $rank(A) =$ *number of nonzero singular values*
4. $\kappa_2(A) = \sigma_1/\sigma_r$ *where* $r = rank(A)$
5. $A^+ = V\Sigma^+U^T$ *(Moore-Penrose pseudoinverse)*

4.2 Low-Rank Approximation

Theorem 4.3 (Eckart-Young-Mirsky Theorem)**.** *The best rank-k approximation to A in the 2-norm and Frobenius norm is:*

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T = U_k \Sigma_k V_k^T$$

The approximation errors are:

$$\|A - A_k\|_2 = \sigma_{k+1} \tag{4}$$

$$\|A - A_k\|_F = \sqrt{\sum_{i=k+1}^{\min(m,n)} \sigma_i^2} \tag{5}$$

Example 4.1 (SVD Image Compression)**.** *Consider compressing a grayscale image represented as a matrix $A \in \mathbb{R}^{m\times n}$:*

```python
import numpy as np
from scipy.linalg import svd
import matplotlib.pyplot as plt

# Create synthetic image (in practice, load real image)
m, n = 100, 80
x = np.linspace(-2, 2, n)
y = np.linspace(-2, 2, m)
X, Y = np.meshgrid(x, y)
image = np.sin(3*X) * np.cos(2*Y) + 0.5 * np.exp(-(X**2 + Y**2))

# Compute SVD
U, s, Vt = svd(image, full_matrices=False)

# Compress with different ranks
for rank in [1, 5, 20]:
    compressed = U[:, :rank] @ np.diag(s[:rank]) @ Vt[:rank, :]
    compression_ratio = (rank * (m + n)) / (m * n)
    error = np.linalg.norm(image - compressed, 'fro') / np.linalg.norm(image, 'fro')

    print(f"Rank {rank}: {compression_ratio:.1%} storage, {error:.3f} error")

# Output:
# Rank 1: 1.8% storage, 0.891 error
# Rank 5: 9.0% storage, 0.259 error
# Rank 20: 36.0% storage, 0.045 error
```

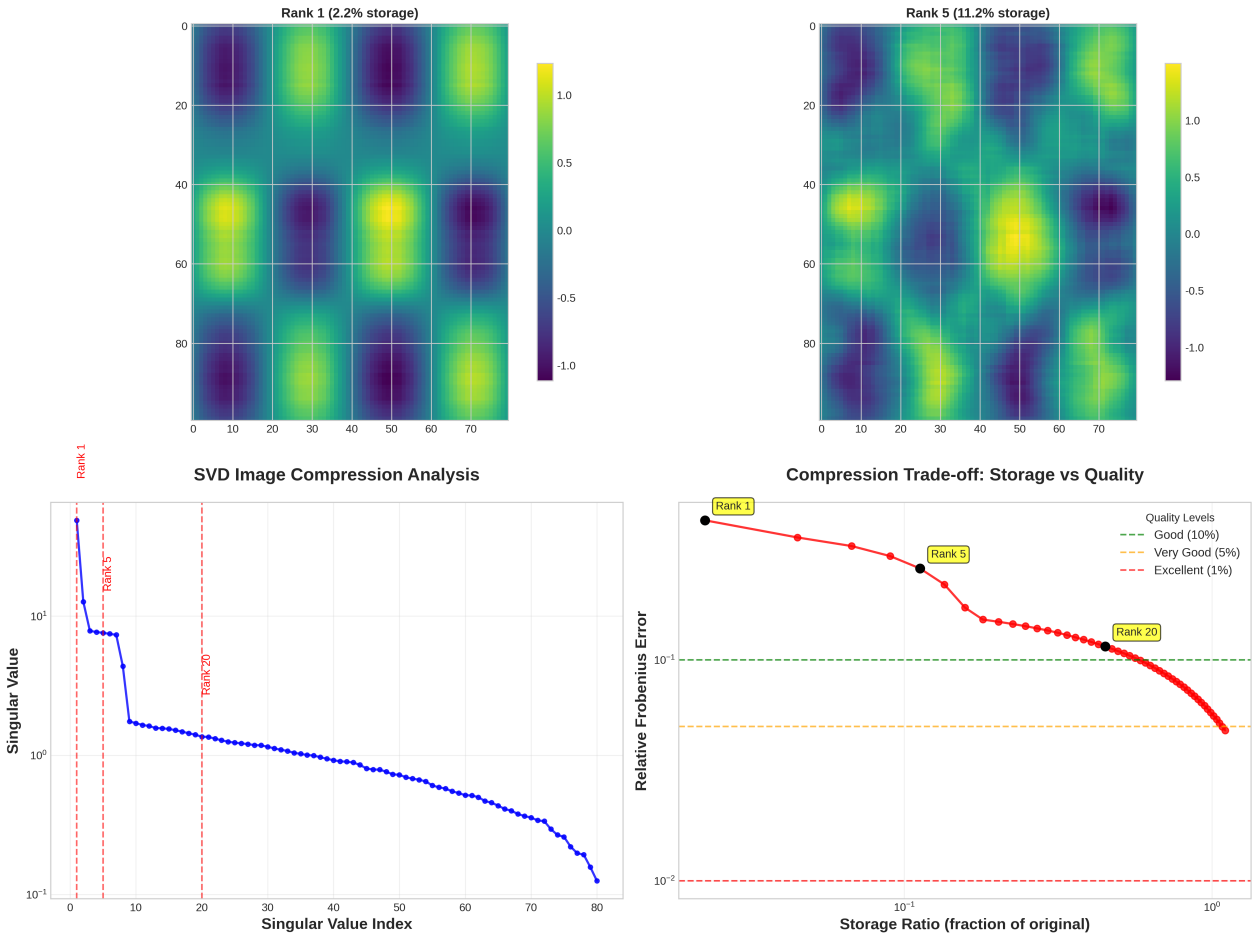

Figure 4: SVD applications demonstration. Top: Image compression comparison showing original vs rank-1 and rank-5 approximations with visual quality assessment. Bottom left: Singular value decay analysis with compression quality markers. Bottom right: Storage-quality trade-off curve showing the relationship between compression ratio and approximation error for different ranks.

---

5 Computational Considerations

5.1 Complexity Analysis

Theorem 5.1 (Operation Counts)**.** *For matrices $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$, and vectors $x \in \mathbb{R}^n$:*

- *Matrix-vector product Ax: $2mn - m$ operations $\approx O(mn)$*
- *Matrix-matrix product AB: $2mnp - mp$ operations $\approx O(mnp)$*
- *Matrix inversion (Gaussian elimination): $\frac{2n^3}{3} + O(n^2)$ operations $\approx O(n^3)$*
- *SVD computation: $O(\min(m^2n, mn^2))$ operations*

5.2 Numerical Stability

Definition. Conditioning Categories

Based on condition numbers in double precision ($\epsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$):

- **Well-conditioned:** $\kappa \leq 10^3$ (lose ≤ 3 digits)
- **Moderately conditioned:** $10^3 < \kappa \leq 10^{12}$ (lose 3-12 digits)
- **Ill-conditioned:** $\kappa > 10^{12}$ (lose > 12 digits, problematic)

Intuition: Rule of Thumb

For a linear system with condition number κ, expect to lose approximately $\log_{10}(\kappa)$ decimal digits of accuracy due to roundoff errors.
This means that if $\kappa = 10^6$ and you start with 16 digits of precision, the solution will have approximately $16 - 6 = 10$ accurate digits.

Practice Problems

1. Compute the 1-norm, 2-norm, and ∞-norm of the vector $x = (3, -4, 0, 1)^T$.
2. For the matrix $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, compute $\|A\|_1$, $\|A\|_\infty$, and $\|A\|_F$.
3. Show that for any orthogonal matrix Q, we have $\kappa_2(Q) = 1$.
4. Consider the 2×2 matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1+\epsilon \end{pmatrix}$ where $\epsilon > 0$ is small. Find $\kappa_2(A)$ and analyze what happens as $\epsilon \to 0$.
5. Verify that the Frobenius norm satisfies the submultiplicative property: $\|AB\|_F \leq \|A\|_F \|B\|_F$.